Great Learning

7-8 minutes

Till now, we've covered what an AI Agent is and the different types of AI Agent architectures.

An AI agent does not exist in isolation. It operates within an environment. The environment is everything outside the agent that

- provides inputs (observations, data, signals)
- is affected by the agent’s actions
- may change over time, with or without the agent’s influence

Examples of environments would include

- A web page an LLM agent is browsing
- A simulation of a game (chess, Pac-Man)
- A company’s internal systems (CRM, email, databases)
- The physical world (for robots, autonomous cars)

In Agentic AI design, understanding the type of environment is as important as understanding the type of agent, because it tells you

- How hard prediction and planning will be
- How much uncertainty the agent must handle
- Whether it must coordinate with others
- How fast it must react to changes

Every agent operates within an environment that defines how it perceives and responds to the world. The structure of this environment, the presence of other agents, and the level of uncertainty involved all influence how the agent thinks and acts.

In this section, we discuss three key ways to understand agent environments:

1. Static vs. Dynamic environments
2. Single-agent vs. Multi-agent environments
3. Deterministic vs. Stochastic environments

1. Static vs. Dynamic Environments

This dimension describes whether the environment changes while the agent is operating.